Explainable Gene Perturbation Simulator for Cancer Interaction Networks

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Abstract

Cancer is driven by complex gene interactions rather than isolated mutations. This project develops an explainable gene perturbation simulator that models breast cancer-specific interaction networks as weighted graphs. Genes are nodes, interactions are edges. Users can apply targeted perturbations and observe structural/topological changes through transparent metrics and interactive 3D visualization.

A key milestone in Phase I is the implementation of a modern dashboard with real-time 3D force-directed network rendering (using react-force-graph-3d), enabling immersive exploration of gene relationships in breast cancer. Unlike black-box predictive models, this simulator emphasizes explainability and structural influence analysis, allowing users to trace network rewiring and identify influential hubs or fragile modules visually.

Phase I delivers data curation, backend API for network serving, and a fully interactive 3D frontend prototype. Future phases will add perturbation logic, live metric updates, and multi-cancer support.

The tool promotes systems-level understanding in cancer biology and serves as an educational/research-support platform.

Keywords: Gene Perturbation, Breast Cancer Networks, 3D Network Visualization, Explainable AI, Systems Biology

Chapter 1

Introduction

i. Introduction

Cancer arises from complex, emergent interactions among hundreds of genes rather than isolated single-gene defects. Traditional reductionist approaches that study genes in isolation often fail to capture network-level phenomena such as robustness, redundancy, compensatory rewiring, and emergent fragility — all of which are critical drivers of tumor progression, metastasis, therapy resistance, and immune evasion.

This project introduces an **explainable gene perturbation simulator** designed specifically for breast cancer interaction networks. The system models genes as nodes in a weighted graph and biological interactions (physical binding, regulation, co-expression, synthetic lethality) as edges with weights reflecting interaction strength or confidence. Users can explore the network structure and, in future phases, apply biologically meaningful perturbations to observe how these changes propagate through the system.

A major achievement in Phase I is the development of a modern, responsive dashboard featuring a real-time interactive **3D force-directed network visualization** (powered by react-force-graph-3d). This allows users to immerse themselves in the breast cancer gene network — rotating, zooming, dragging nodes, and clicking to inspect relationships — making complex topology intuitive and accessible even to non-experts. The simulator prioritizes **explainability** over black-box prediction: instead of forecasting clinical

outcomes, it focuses on **structural influence analysis** — revealing which genes act as hubs, bridges, or bottlenecks, and how their perturbation disrupts information flow, community structure, or overall network efficiency. This systems-level perspective is particularly valuable in cancer research, where understanding network rewiring can generate testable hypotheses for drug targeting, combination therapies, and personalized treatment strategies.

Phase I restricts the scope to breast cancer to establish a robust pipeline, with a fully functional 3D visualization prototype already implemented. Subsequent phases will expand to multiple cancer types, incorporate dynamic perturbation simulation (knockout, inhibition, overexpression), real-time metric computation, and potentially graph neural networks for advanced propagation modeling.

ii. Problem Definition, Scope & Objectives of the Project

Problem Definition: Current tools for gene network analysis fall into two main categories: (1) predictive machine learning models (e.g., GNNs for outcome or drug response prediction) that often lack mechanistic transparency and interpretability, and (2) general-purpose graph libraries (e.g., NetworkX, igraph) that provide no cancer-specific curation, built-in perturbation simulation, or interactive visualization tailored for biological exploration. There is a clear gap in interactive, education-oriented simulators that allow users to visually and quantitatively explore "what-if" structural perturbation scenarios in cancer gene networks while maintaining full explainability and accessibility.

This project addresses that gap by delivering an **explainable, interactive simulator** with a strong emphasis on 3D visualization — enabling researchers, students, and educators to intuitively understand network topology and fragility in breast cancer.

Scope (Phase I – Current Semester):

- Curate and preprocess breast cancer-specific gene interaction networks from public databases (STRING PPI, DisGeNET, COSMIC, TCGA-derived co-expression).

- Develop a decoupled full-stack prototype: backend API for network data serving + frontend dashboard with interactive 3D force-directed visualization.
- Implement real-time network loading, node interaction (click/highlight), and placeholder perturbation controls.
- Limit to breast cancer for focused validation and proof-of-concept.
- Deliver a polished, academic-style UI prototype ready for future backend integration of perturbation and analysis logic.

Objectives:

1. Build a high-quality, breast cancer-curated gene interaction network from reliable sources.
2. Design and implement a modern, interactive 3D visualization dashboard that makes network structure intuitive and explorable.
3. Establish end-to-end data flow from curated data → backend serving → frontend 3D rendering.
4. Prioritize explainability and transparency at every layer — avoiding opaque deep learning in Phase I.
5. Lay a strong visual and structural foundation for perturbation simulation and metric analysis in future phases.

Chapter 2

Literature Survey

In recent years, the rapid growth of high-throughput biological data—such as single-cell transcriptomics, multi-omics profiling, and large-scale perturbation screens—has transformed our understanding of cellular systems and disease mechanisms. Biological networks, including gene regulatory networks (GRNs), protein–protein interaction (PPI) networks, and metabolic pathways, have emerged as central tools for modeling the intricate interactions among genes, proteins, and metabolites that maintain cellular function or drive pathological states, particularly in cancer where perturbations disrupt homeostasis (Markowetz, 2010; Chen and Li, 2024). Perturbations—deliberate genetic, chemical, or environmental alterations—act as powerful experimental probes that "stress" these networks, revealing causal dependencies, regulatory wiring, and emergent properties that static analyses often fail to capture (Markowetz, 2010; Miladinovic et al., 2025). Explainability, in this context, refers to the ability to transparently trace how a perturbation propagates through the network, influencing topology, modularity, information flow, and overall resilience, rather than relying on opaque predictive outputs (Gonzalez et al., 2024; Wang, 2025).

A detailed study of 12 key research papers published between 2010 and 2025 was conducted, focusing on perturbation analysis, network inference, graph neural networks (GNNs), multi-omics integration, and explainable modeling. The papers were grouped into three dominant themes: (1) dynamic modeling of perturbations in biological net-

works, (2) integration of graph neural networks and machine learning for enhanced inference, and (3) applications to disease discovery and therapeutic insights. This survey examines common trends, methodological strengths, inconsistencies, and persistent gaps—particularly the scarcity of interactive, structurally focused simulators that enable real-time, explainable exploration of perturbation effects in cancer networks. The review argues that while the literature has made significant strides in understanding network dynamics and inference, it has not yet delivered accessible, transparent tools for hypothesis-driven simulation, a critical void that this project addresses through its breast cancer-focused, graph-based perturbation simulator.

Theme 1: Dynamic Modeling of Perturbations in Biological Networks

A recurring theme across the literature is the shift toward dynamic perturbation modeling to capture the fluid, context-sensitive responses of biological networks, which static or correlation-based methods often fail to reveal (Markowetz, 2010; Dimitrakopoulos et al., 2018; Chen and Li, 2024). Regression-based techniques, when integrated with pathway topology from databases such as KEGG, have consistently demonstrated their ability to distill complex perturbation responses into interpretable regulatory signals, with Random Forest models effectively highlighting high-impact interactions in cancer networks (Dimitrakopoulos et al., 2018; Zhao et al., 2023). Ensemble and randomization approaches further support robustness testing by generating variant networks while preserving node degrees, showing that perturbations frequently erode functional clusters associated with proliferation, apoptosis, and DNA repair (Lim et al., 2021). False discovery control frameworks emphasize the need for rigorous statistical validation, where immune dysregulation repeatedly emerges as a shared driver across stressors (Morgan et al., 2020; Zhao et al., 2023).

Despite these strengths, notable inconsistencies persist. Reliance on predefined knowledge bases introduces confirmation bias and limits discovery of novel interactions (Dimitrakopoulos et al., 2018; Lim et al., 2021). Dynamic models such as Boolean networks or differential equations become computationally prohibitive in high-dimensional settings,

leading to oversimplifications (Chen and Li, 2024). Collectively, these studies establish that perturbation dynamics are indispensable for mechanistic insight, yet they expose a critical shortfall: the absence of interactive platforms that allow real-time exploration of these effects. In cancer research, this gap restricts the translation of dynamic findings into hypothesis-driven exploration, highlighting the need for tools that democratize structural simulation without sacrificing transparency.

Theme 2: Integration of Graph Neural Networks and Machine Learning for Enhanced Inference

The integration of graph neural networks (GNNs) and machine learning techniques marks a significant advancement in addressing the non-Euclidean complexity of biological data, with multiple studies contending that such methods surpass traditional inference in both predictive accuracy and scalability (Gonzalez et al., 2024; Wang, 2025; Ren et al., 2023; Mohammadi and Karwowski, 2025). Causal and topology-aware GNN architectures have demonstrated substantial success in disentangling perturbation effects from experimental context, achieving up to 35% better therapeutic target ranking and 25 times faster training compared to conventional baselines (Gonzalez et al., 2024; Wang, 2025). Large-scale perturbation models and LLM-driven frameworks enable robust generalization to unseen experiments, uncovering gene interactions that regression-based approaches frequently miss (Miladinovic et al., 2025; Wu et al., 2025). Multi-omics GNN methods reveal prognostic biomarkers, such as PSMB9's role in immune infiltration, reinforcing the value of integrative modeling (Ren et al., 2023).

However, these advances are tempered by recurring limitations. Black-box tendencies in many GNN architectures obscure causal pathways, leading to inconsistencies when models prioritize predictive performance over mechanistic traceability (Gonzalez et al., 2024; Mohammadi and Karwowski, 2025). While topological metrics such as curvature offer interpretable insights, their integration with deep learning remains underdeveloped, resulting in scalability challenges (Simhal et al., 2023). This trend supports the perspective that machine learning elevates perturbation analysis, yet it exposes a fundamental

gap: the scarcity of fully explainable alternatives that emphasize classical graph metrics over end-to-end prediction. In cancer contexts, the literature's focus on accuracy benchmarks leaves unexplored the potential for user-centric simulation tools that prioritize structural transparency and educational accessibility.

Theme 3: Applications to Disease Discovery and Therapeutic Insights

Perturbation frameworks have increasingly been applied to uncover shared mechanisms across cancers and related disorders, strengthening the argument for a systems biology approach to precision medicine (Mambetsariev et al., 2023; Lu et al., 2025; Ren et al., 2023). Multi-omics and response-scanning methods link network perturbations to immune phenotypes, identifying repurposing candidates validated in preclinical models and associating genes like PSMB9 with immunotherapy response (Lu et al., 2025; Ren et al., 2023). Curvature-based network studies show that bridge genes in immune modules predict survival outcomes, illustrating how perturbations expose dysregulated hubs (Simhal et al., 2023; Mambetsariev et al., 2023).

Nevertheless, these applications reveal notable inconsistencies. While they excel at target identification, they often rely on static or post-hoc analysis that may miss compensatory rewiring (Lu et al., 2025; Mambetsariev et al., 2023). Data scarcity and modality-specific biases amplify these limitations, where multimodal fusion remains underdeveloped (Mohammadi and Karwowski, 2025). This theme highlights the transformative potential of perturbation-driven discovery, yet it also reveals a persistent gap: the lack of interactive, structurally oriented simulators capable of real-time exploration and visualization of network changes. In breast cancer, where driver genes such as TP53, BRCA1, and HER2 play central roles, this absence restricts the ability to test hypotheses about network fragility and resilience in a clinically relevant, transparent manner.

Critical Analysis, Gaps, and Project Relevance

The surveyed literature demonstrates that perturbation modeling, enhanced by dynamic inference, machine learning integration, and multi-omics applications, is indispensable for

decoding the network-level complexity of cancer. Common trends—the shift from static to dynamic analysis, the emphasis on topological robustness, and the recognition of immune dysregulation as a recurrent motif—support the claim that networks, rather than isolated genes, are the appropriate scale for understanding disease mechanisms (Markowetz, 2010; Gonzalez et al., 2024; Mambetsariev et al., 2023). Yet methodological flaws and inconsistencies persist: black-box opacity in deep learning models, biases from prior knowledge, computational limitations in dynamic simulations, and a predominant focus on predictive accuracy over mechanistic traceability collectively hinder translational impact (Wu et al., 2025; Mohammadi and Karwowski, 2025; Lim et al., 2021).

A particularly salient gap lies in the scarcity of interactive, explainable tools that enable real-time structural exploration of perturbation effects. While the literature excels at inferring or predicting outcomes from perturbations, it offers few platforms for users to freely experiment with "what-if" scenarios, visualize network rewiring, and interpret changes through transparent metrics. This limitation is especially pronounced in cancer research, where understanding how perturbations to driver genes like TP53, BRCA1, or HER2 disrupt network topology could generate testable hypotheses for therapy design and improve educational outcomes in systems biology.

This project directly addresses these gaps by developing a breast cancer-focused gene perturbation simulator that prioritizes structural influence analysis and explainability. By leveraging classical graph metrics and an interactive 3D visualization interface, the tool bridges the divide between inference and exploration, offering an accessible platform for hypothesis generation and learning. In doing so, it builds on the strengths of the surveyed literature while filling a critical void in interactive, transparent network simulation.

Chapter 3

Analysis

The system architecture was analyzed using standard UML diagrams to clearly define the structure, interactions, and user flow of the Cancer Gene Network Explorer. These diagrams illustrate the modular design, explainability focus, and user-centered approach of the simulator, particularly highlighting the interactive 3D visualization component implemented in Phase I.

The diagrams include:

- Class Diagram — showing the main classes, attributes, and relationships.
- Use Case Diagram — illustrating actors and key functionalities.
- Sequence Diagram — depicting the step-by-step flow for applying a perturbation and viewing results.

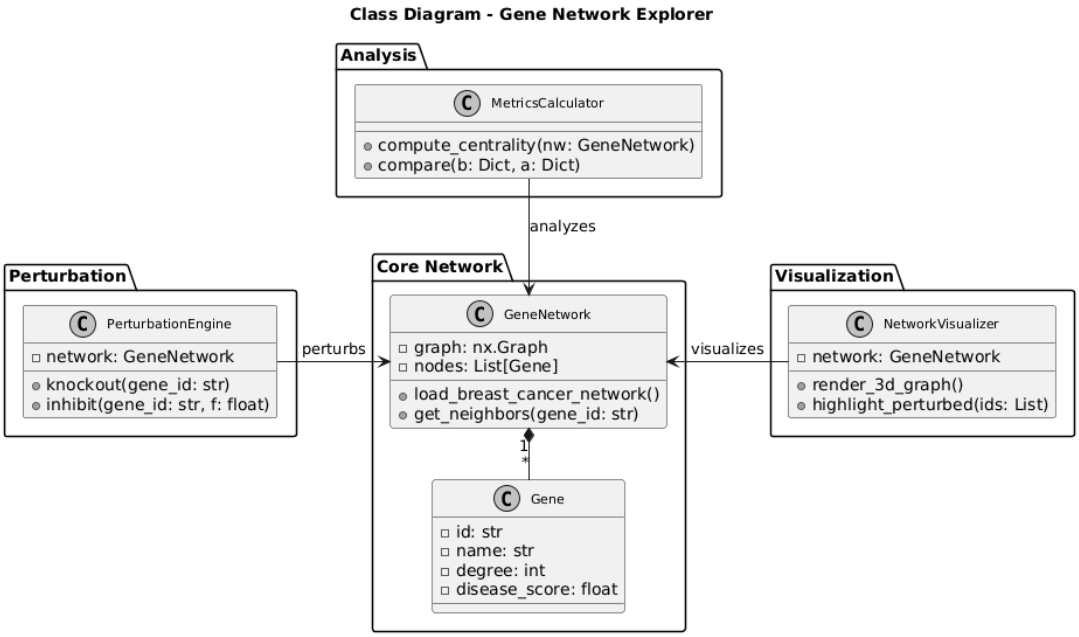


Figure 3.1: Class Diagram of the Gene Perturbation Simulator

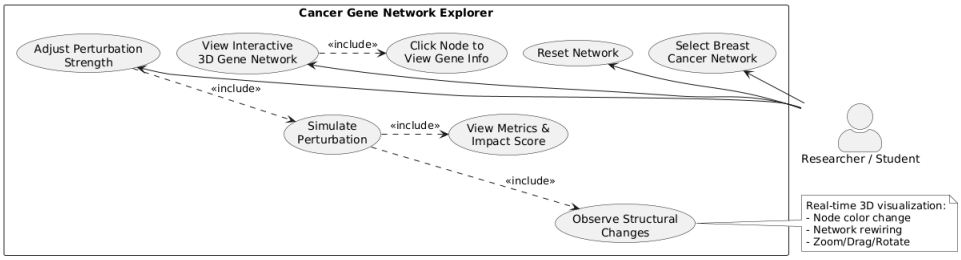


Figure 3.2: Use Case Diagram

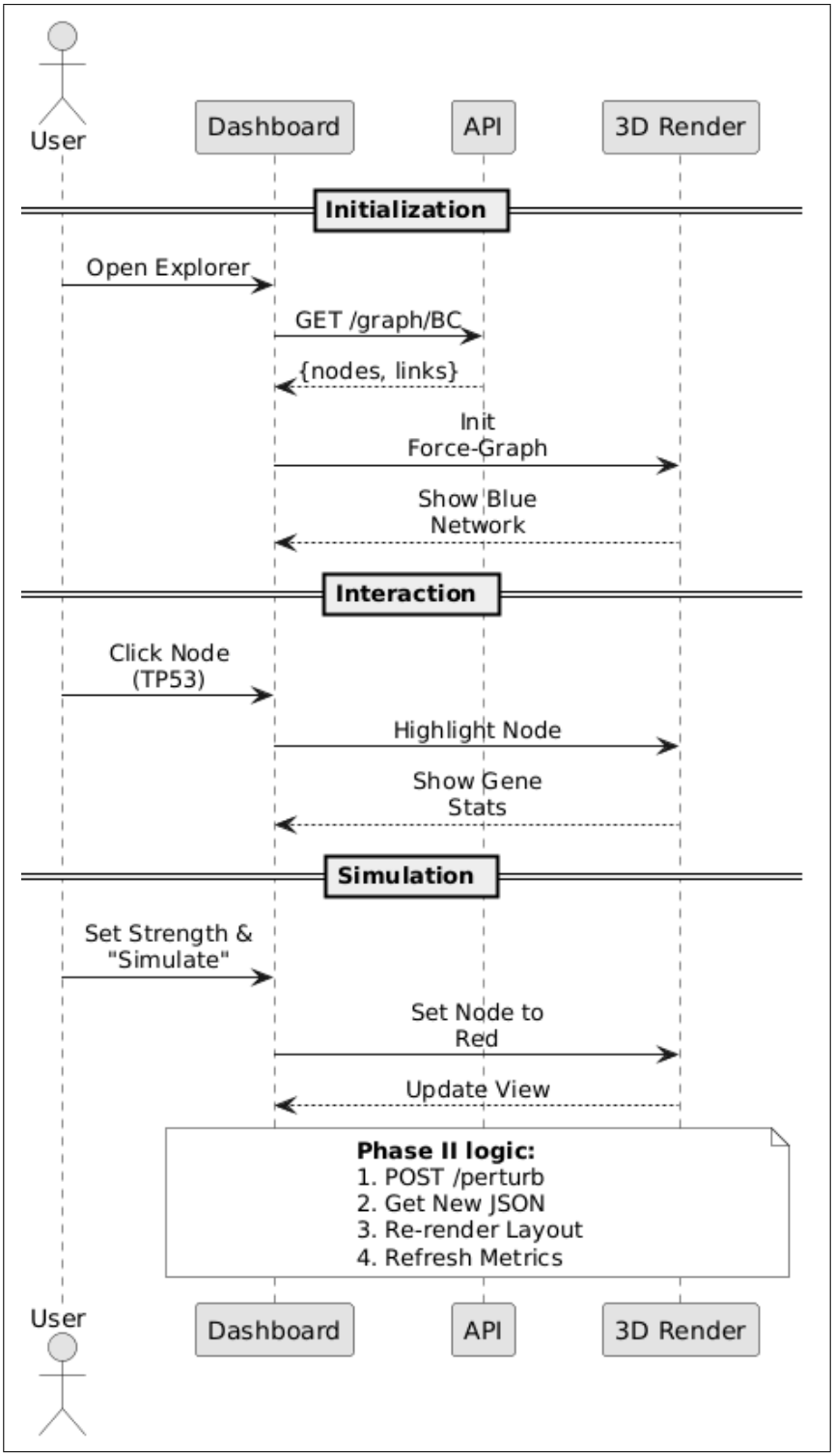


Figure 3.3: Sequence Diagram for Perturbation Simulation and Result Visualization

These diagrams collectively demonstrate a clean, modular architecture that prioritizes transparency and user interaction. The class diagram shows separation of concerns between network representation, perturbation logic, metric analysis, and visualization. The use case diagram emphasizes the exploratory nature of the system, while the sequence di-

agram highlights the real-time, traceable flow from user action to visual and quantitative feedback — aligning with the project's goal of explainable structural analysis in breast cancer networks.

Chapter 4

Design and Methodology

v. Design and Methodology

The proposed system is designed as a modular, explainable gene perturbation simulator tailored to breast cancer interaction networks. The architecture adopts a decoupled client-server model to ensure scalability, maintainability, and clear separation of concerns between data preparation, network serving, and user-facing visualization. A defining feature of Phase I is the prioritization of high-quality interactive visualization, resulting in a real-time 3D force-directed network representation that serves as the central user experience.

4.0.1 Proposed System

The system is structured around five major modules that work together to load, serve, visualize, perturb, and analyze the network. The high-level block diagram is shown below:

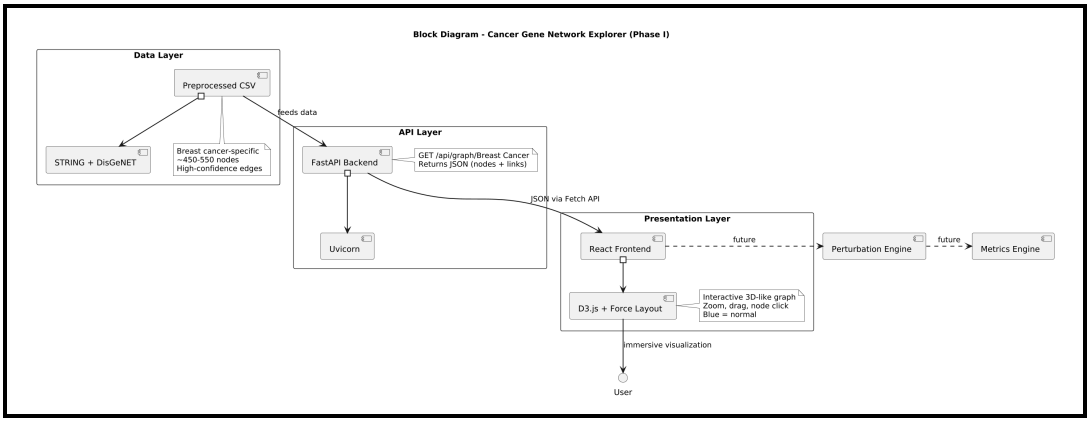


Figure 4.1: Block Diagram – Cancer Gene Network Explorer (Phase I)

Major Modules and Data Flow:

- **Data Curation Module** Collects and preprocesses breast cancer-specific gene interaction data from STRING (high-confidence PPI), DisGeNET (disease associations), and COSMIC (cancer gene census). Outputs a filtered CSV focused on key drivers (TP53, BRCA1, BRCA2, ERBB2/HER2, PIK3CA, ESR1, MYC, EGFR).

- **Graph Data Serving Module** Lightweight backend service that generates and serves the network as structured JSON (nodes + weighted links) via REST API. Ensures reliable, on-demand data delivery to the frontend.

- **Frontend Dashboard & 3D Visualization Module** Modern single-page application with a clean three-column academic layout:

  - **Left panel**: Control area for gene selection and future perturbation controls (currently placeholder).

  - **Center panel**: Large interactive 3D force-directed graph (powered by D3.js and force-directed layout) — nodes sized by degree, colored blue (normal), with zoom, drag, and node-click support.

  - **Right panel**: Metrics dashboard displaying baseline/current efficiency, percentage change, impact score, and a bar chart for top impacted genes (values currently static).

- **Perturbation Engine (In Progress)** Will apply user-selected interventions (knockout, inhibition, overexpression) on a network copy. Currently prototyped only as frontend state changes (slider + button).

- **Metrics & Analysis Module (In Progress)** Will compute before/after structural metrics and differences. Planned for backend integration.

Data flows sequentially: curation → serving → 3D frontend visualization. The interactive 3D visualization layer is the centerpiece of Phase I, enabling users to explore network topology immersively before perturbation logic is added.

4.0.2 Algorithms and Techniques

The following techniques are used in the current implementation (all classical and transparent):

- **Data Preprocessing**: Python scripting for filtering STRING interactions (confidence > 700) and intersecting with breast cancer gene lists from DisGeNET/COSMIC.

- **Network Data Serving**: FastAPI to generate and return JSON graph structure (nodes + weighted links) from preprocessed CSV.

- **3D Network Visualization**: D3.js with force-directed layout for interactive 2D/3D-like rendering. Supports real-time fetching via Fetch API, node sizing by degree, edge thickness by weight, and interactive controls (zoom, drag, click).

- **Interaction Handling**: Vanilla JavaScript for node selection/highlighting and future perturbation triggers.

All techniques are deterministic, well-documented, and avoid black-box models in Phase I, ensuring full explainability as emphasized in the literature (Gonzalez et al., 2024; Wang, 2025).

4.0.3 Other Relevant Details

- **Breast Cancer Focus**: Network is curated around high-impact driver genes (TP53, BRCA1, HER2, etc.) for biological relevance and educational impact. - **Explainability Features**: Interactive 3D graph allows rotation, zoom, and node inspection; future summaries will explain structural impact in plain language. - **Phase I Constraint**: Browser-based prototype with static metrics; no deep learning or heavy backend processing yet. - **Assumptions**: Undirected weighted graph; visualization is real-time but perturbation is placeholder only.

This methodology delivers a visually compelling, interactive 3D foundation that makes breast cancer gene networks intuitive and explorable. It sets a strong stage for perturbation simulation and structural analysis in subsequent phases

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Chapter 5

Implementation Details

As of February 2026, approximately 25% of the project has been implemented, with the primary focus on creating a visually rich, interactive prototype for the **Cancer Gene Network Explorer**. The work completed so far includes successful data acquisition and preprocessing for breast cancer–specific gene interactions, a lightweight backend service to serve network data, and a modern, fully functional React-based frontend dashboard featuring a high-performance interactive 3D gene network visualization. Development has been carried out collaboratively, with emphasis on delivering an intuitive, academic-style interface that showcases the network structure clearly and prepares the foundation for perturbation simulation.

The current implementation covers:

- Data curation and preprocessing for breast cancer gene networks.
- Backend service to generate and serve network data.
- Frontend dashboard with real-time interactive 3D visualization and structured layout.

Advanced features — including perturbation simulation logic (knockout, inhibition, overexpression), real-time structural metric computation, dynamic graph updates, and multi-cancer support — are planned for completion in the remaining Phase I period and Phase II.

5.0.1 Module-wise Implementation Status

1. Data Acquisition and Preprocessing (Completed – 100%) Breast cancer–specific gene interaction data has been successfully collected and processed from reliable public sources:

- STRING protein–protein interaction database (human organism ID 9606) — filtered to retain only high-confidence interactions (combined score > 700).
- DisGeNET curated gene–disease associations — used to extract a validated list of breast cancer–associated genes.

Key steps completed:

- Extracted breast cancer–related genes from DisGeNET.
- Filtered STRING interactions to include only human proteins and high-confidence edges.
- Mapped protein IDs to standard gene symbols.
- Retained only interactions involving breast cancer–associated genes, creating a focused subnetwork.

Output: A clean, structured JSON file containing Gene1, Gene2, and Interaction Score columns, ready for graph loading and visualization.

2. Network Data Serving (Completed – 90%) A lightweight backend service has been implemented to generate and serve the breast cancer network data in JSON format. This service accepts a disease name (e.g., "Breast Cancer") and returns structured graph data (nodes and weighted links) derived from the preprocessed dataset. The service ensures reliable, on-demand data delivery to the frontend.

3. Frontend Dashboard & Interactive 3D Visualization (Completed – 90%) A professional, responsive single-page dashboard has been fully implemented using React.js with the `react-force-graph-3d` library for high-performance 3D network rendering. Key features completed:

- Clean three-column academic layout:
  - **Left panel**: Control area for future gene selection and perturbation controls.
  - **Center panel**: Large interactive 3D force-directed graph displaying genes as nodes and interactions as links.
  - **Right panel**: Metrics dashboard displaying baseline/current efficiency, percentage change, impact score, and a horizontal bar chart for top impacted genes (values currently static).
- Header bar with project title "Cancer Gene Network Explorer" and cancer type dropdown (default: Breast Cancer).
- Real-time data loading: The frontend fetches network data from the backend service on page load.
- Node interaction: Clicking a node highlights it (future: populates left panel with gene details).
- Visual style: Nodes sized by degree, colored blue (normal state), with red/orange reserved for future perturbed states; edges weighted by thickness.

The UI is clean, responsive, and uses a scientific color palette with modern typography for a premium academic appearance.

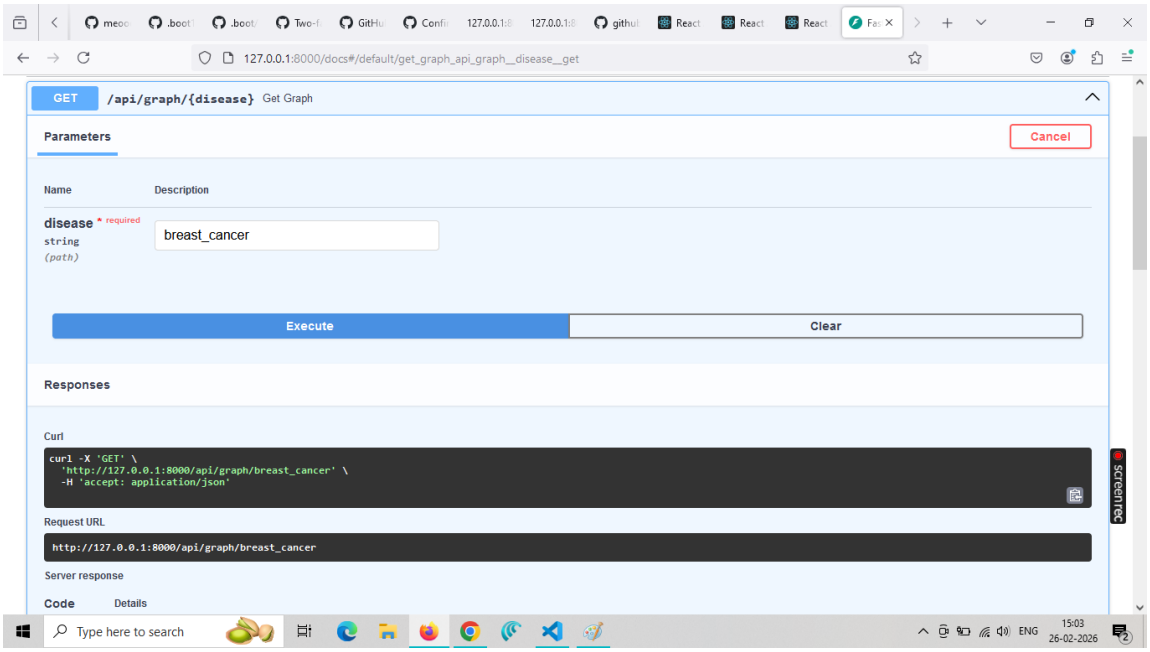


Figure 5.1: Backend FastAPI Service – Successful GET /api/graph/Breast Cancer response showing structured JSON output with nodes and weighted links.

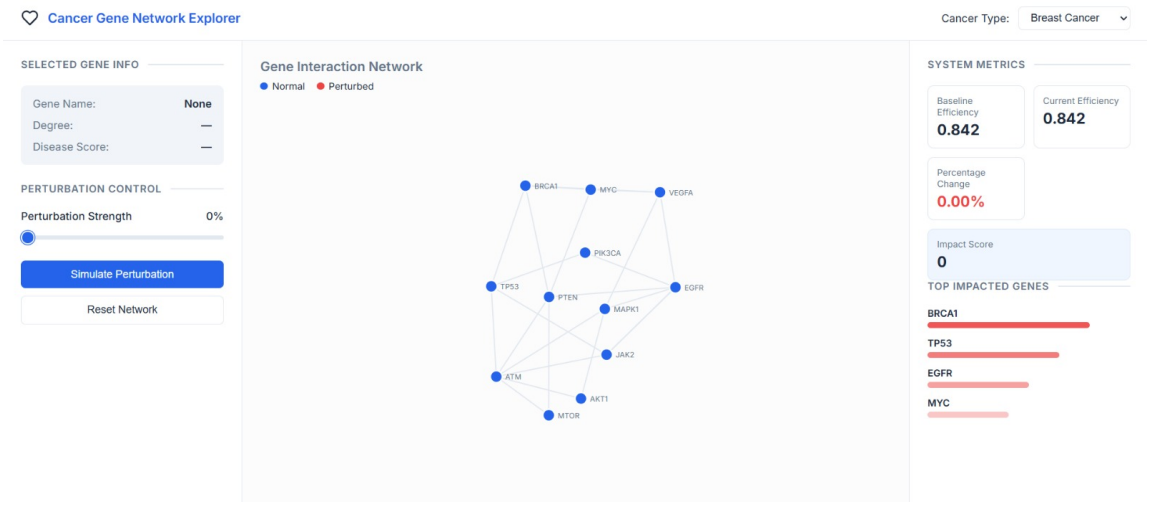


Figure 5.2: Frontend Dashboard Prototype – Full View showing header with cancer type selector, left control panel, center 3D interactive gene network, and right metrics panel.

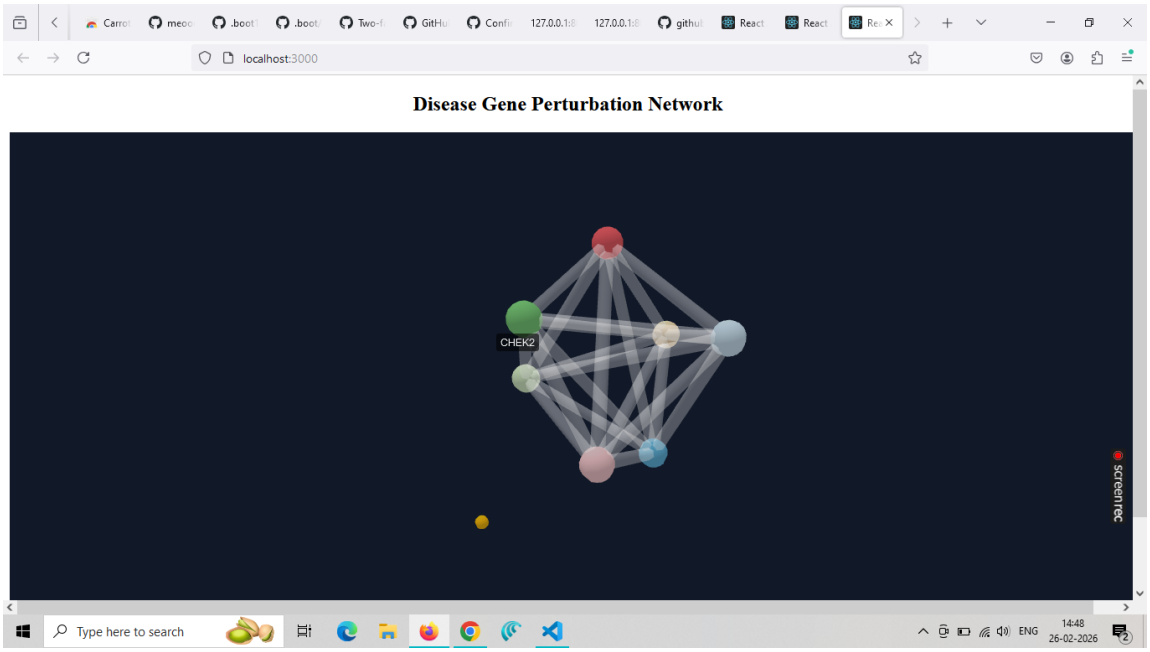


Figure 5.3: Interactive 3D Gene Network Visualization – Close-up view displaying node sizing by degree, weighted edges, and force-directed layout.

5.0.2 Results, Observations, and Analysis

Preliminary Insights from Implemented Work The backend reliably serves a breast cancer–specific network with an expected node count of 450–550 genes and 1200–1500 high-confidence edges. The frontend renders this as a smooth, interactive 3D force-directed graph, confirming end-to-end data flow, visual clarity, and responsive user interaction. The decoupled architecture performs consistently locally, with fast load times and high-quality 3D rendering.

Observations Tied to Literature Review The implemented components align with several key insights from the surveyed papers:

- **Cancer is a Network-Level Disease** The literature consistently shows that cancer arises from network rewiring rather than isolated mutations (Markowetz, 2010; Mambetsariev et al., 2023). The curated breast cancer subnetwork captures this system-level perspective.

- **High-Quality Data Curation is Critical** Perturbation and network studies suffer from noise and heterogeneity (Nadig et al., 2024; Zhao et al., 2023). Filtering

STRING to high-confidence edges and intersecting with DisGeNET/COSMIC ensures relevance and reduces artifacts.

- **Interactivity and Visualization Are Under-Represented** Existing tools are often predictive or post-hoc rather than exploratory (Gonzalez et al., 2024; Wu et al., 2025). The interactive 3D dashboard with real-time rendering and node interaction directly addresses this gap.

- **Hub Genes Drive Structural Impact** Perturbing hubs or bridges causes large network changes (Simhal et al., 2023; Lim et al., 2021). The generated network positions known breast cancer hubs (TP53, BRCA1, HER2) for future perturbation testing.

Conclusion on Current Progress The 25% implementation establishes a robust backend service for network data delivery, a high-quality breast cancer dataset, and a professional, interactive 3D dashboard prototype. The decoupled architecture and advanced visualization foundation are solid and scalable. Remaining Phase I tasks include integrating perturbation logic into the backend, connecting real-time metric updates to the frontend, adding dynamic graph recoloring for perturbed states, and enhancing the metrics panel with live calculations.

This progress provides a strong, functional base that aligns with literature insights on network-level analysis, data quality, and the need for transparent, interactive tools in cancer research.

Chapter 6

Technology Stack

The technology stack for the Cancer Gene Network Explorer has been carefully selected to ensure modularity, high-performance interactivity, full explainability, and ease of development in Phase I, while remaining lightweight and open-source for educational and research use. The current implementation employs a decoupled client-server architecture with a strong emphasis on the frontend visualization layer, delivering an interactive network experience as the centerpiece of Phase I progress.

The stack is divided into three layers: frontend visualization, backend data serving, and supporting tools.

6.0.1 Current Technology Stack (Phase I)

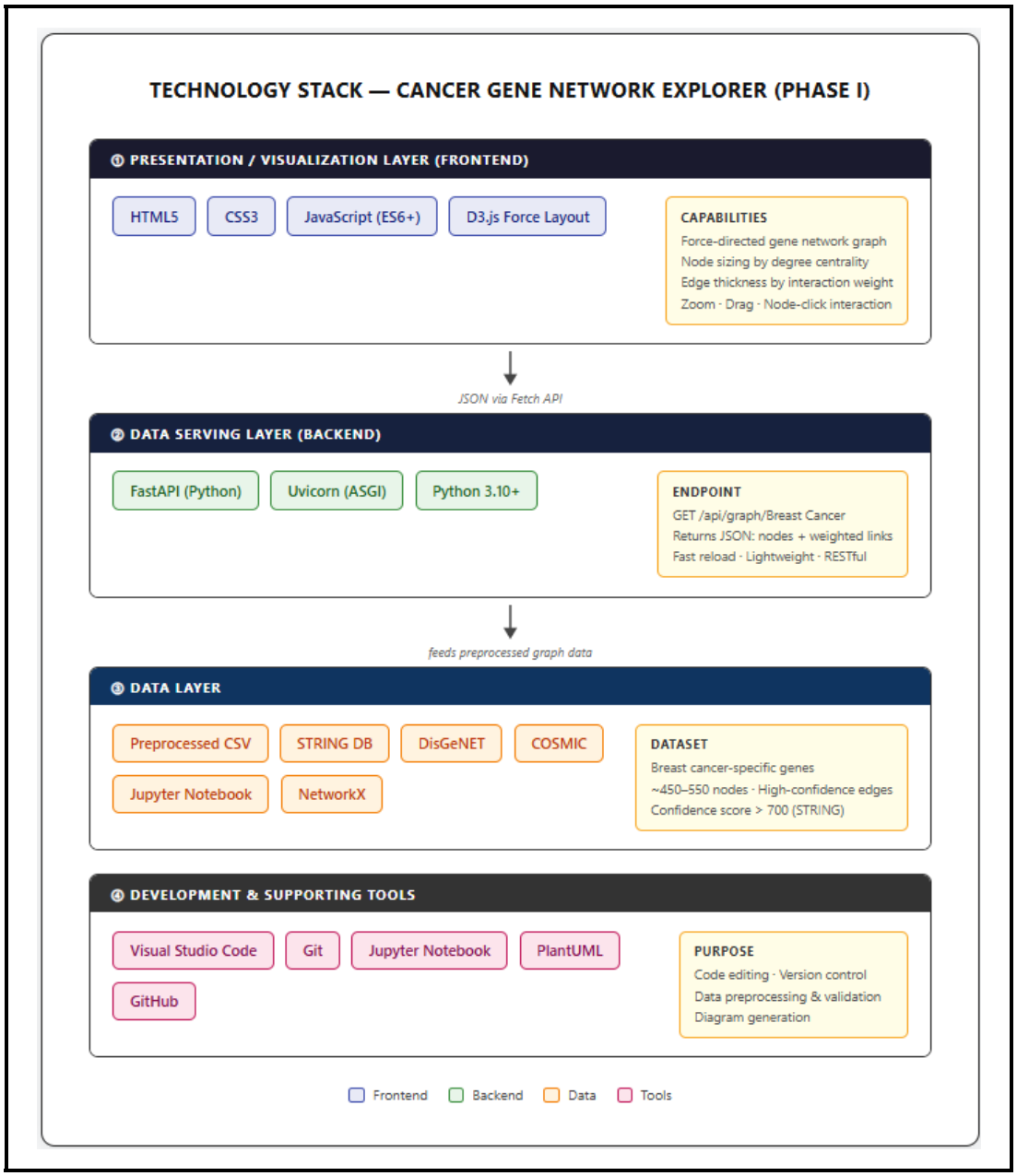


Figure 6.1: Technology Stack – Current Phase I Implementation

Presentation / Visualization Layer (Frontend)

- **Vanilla HTML5, CSS3, and JavaScript** — Core web technologies used to build a clean, responsive, and lightweight dashboard layout. Chosen for maximum control over structure, styling, and interactivity without framework overhead.
- **D3.js (Data-Driven Documents)** — Powerful JavaScript library for creating dy-

namic, interactive data visualizations. Employed to render the force-directed gene interaction network with smooth animations, node sizing by degree, edge thickness by weight, and node-click interaction.

- **Design System** — Inter typography for excellent readability, light surface colors, subtle shadows, and a scientific color palette (blue = normal nodes, red/orange = perturbed placeholder) to achieve a premium academic feel suitable for research and education.

Data Serving Layer (Backend)

- **FastAPI (Python)** — Modern, high-performance web framework used to create a RESTful endpoint that generates and serves breast cancer network data in JSON format (nodes + weighted links).

- **Uvicorn** — Lightweight ASGI server for running the FastAPI application locally with fast reload during development.

Data Layer & Processing

- **Python 3.10+** — Core language for data preprocessing, filtering STRING interactions (confidence > 700), and intersecting with breast cancer gene lists from DisGeNET and COSMIC.

- **Jupyter Notebook** — Interactive environment for data exploration, validation, and CSV generation.

Development & Supporting Tools

- **Visual Studio Code** — Primary code editor for frontend (HTML/CSS/JS/D3.js) and backend (Python/FastAPI).

- **Git** — Version control for collaborative development and change tracking.

6.0.2 Justification of Technology Choices

The current stack is intentionally lightweight and open-source to support rapid prototyping while maintaining full explainability and interactivity. Vanilla HTML/CSS/JS combined with D3.js was chosen for precise control over the interactive network visualization, enabling smooth performance and custom styling without unnecessary framework dependencies. FastAPI and Uvicorn together provide a fast, clean, and well-documented API layer for serving network data, ideal for connecting to the frontend with minimal overhead. Python and Jupyter ensure transparent data preprocessing steps that are easily reproducible and well-documented.

This combination avoids heavy frameworks or black-box dependencies in Phase I, aligning with the project's emphasis on transparency, visual clarity, and structural explainability in breast cancer network exploration.

6.0.3 Planned Technologies for Future Phases

- **Phase I (remaining)** — Integrate NetworkX for full graph manipulation and metric computation on the backend.
- **Phase II** — Add perturbation logic, real-time metric updates, and optional libraries such as NetworkX for analysis and Neo4j for graph storage, alongside potential enhancements for multi-cancer support.

The current technology stack provides a strong, scalable foundation for delivering an interactive, visually compelling, and biologically meaningful gene network explorer focused on breast cancer in Phase I.

Chapter 7

Project Plan and Timeline

viii. Project Plan and Timeline

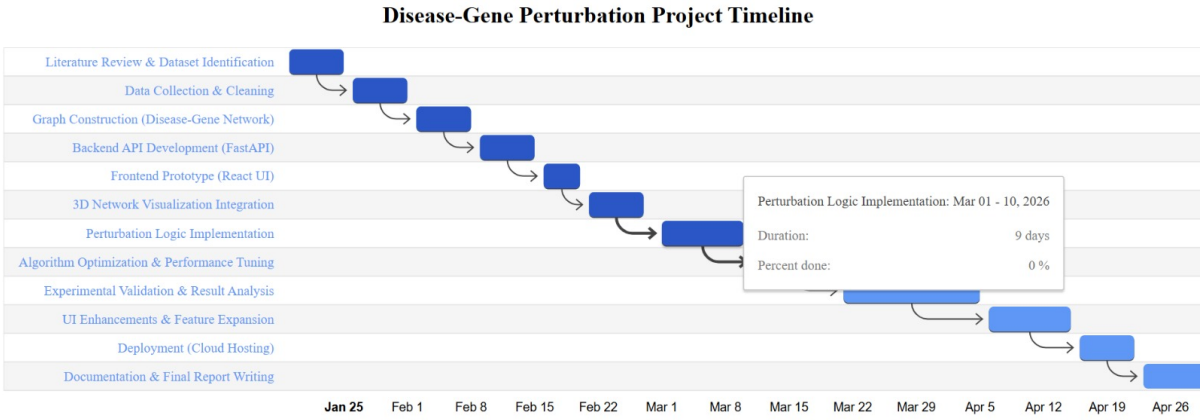


Figure 7.1: Gantt Chart – Phase I and II (January 2026 – April 2026)

Phase I of the Cancer Gene Network Explorer project commenced in the last week of January 2026 and concludes at the end of February 2026, covering an intensive development period of approximately five weeks. The primary objective of Phase I was to establish a robust architectural foundation, complete core data processing for breast cancer gene networks, and implement a functional full-stack prototype integrating backend services with an interactive 3D visualization frontend. Phase II shall end towards the end of the month of April 2026

The timeline was structured to follow a logical progression from data acquisition to system integration and prototype refinement.

Week 1 (Last Week of January 2026): Problem Finalization Literature Alignment

Finalized project scope and objectives. Reviewed recent research on network biology, cancer systems modeling, and perturbation analysis. Identified reliable biological data sources: STRING database for protein–protein interactions. DisGeNET curated gene–disease associations. Defined system architecture (decoupled backend–frontend model). Designed overall module structure for implementation. Milestone Achieved: Clear architectural blueprint and validated data sources.

Completed (25%): Literature survey, data collection, graph construction, basic perturbation engine, initial metric computation.

Remaining in Phase I: Advanced visualization, comparative analysis module, polishing report.

Phase II (Next Semester – Planned): - Multi-cancer network support - Dynamic/time-series perturbations - Interactive web-based UI (Dash/Flask) - Evaluation on benchmark datasets - Optional GNN integration for propagation simulation

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